

Catharsis: An Emotion Driven Music Generator

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**5C34: Advanced AI
Final Presentation**

1 April 2026



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The University of Dublin

Adding more features

After our last presentation we decided to include a few more features and make the project even better using the feedback given to us. Each of us worked on a separate element of the music generator which catapults our project into being more inclusive and deployment ready. Basically you get more outta it.

Ishan

Emotion Conditioned
Music Generation With
Semantic Alignment

Haoran

Multilingual support

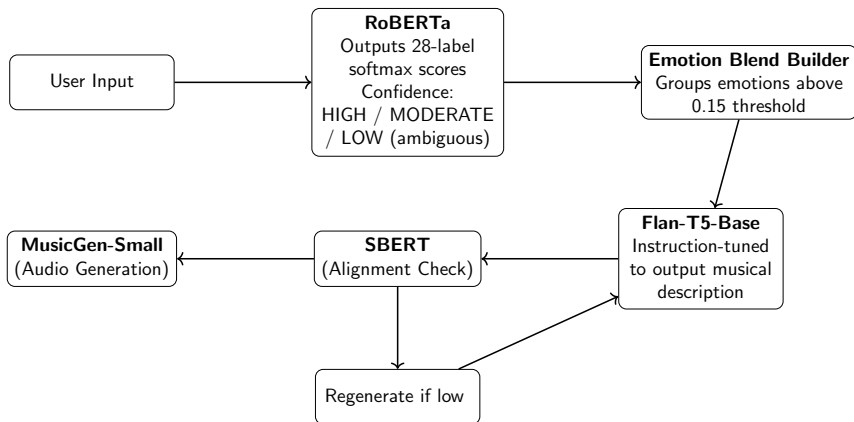
Alba

Multimodal Cinematic
Scene-to-Score Generation
with Audio and Symbolic
Representation

Architecture Upgrade

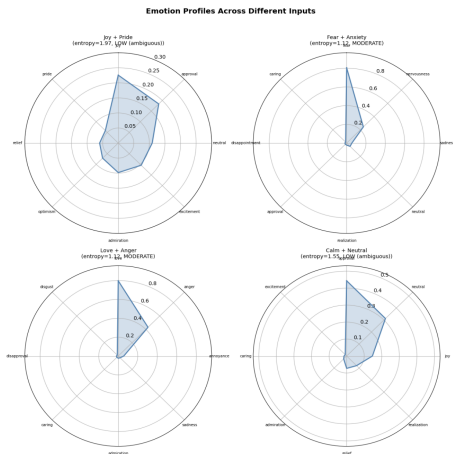
Component	Preview	Draft3	Impact
Emotion Classifier	DistilBERT (6 labels)	RoBERTa + GoEmotions (28 labels)	4.7× more emotional granularity
Prompt Generator	Fixed dictionary	Flan-T5-Base LLM (250M params)	Unlimited unique prompts, instruction-tuned
Semantic Alignment	None	SBERT + feedback loop	Automatic retry if semantic mismatch
Audio Evaluation	None	FAD metric infrastructure	Quantitative quality measurement
Total Models	2	4	More sophisticated pipeline

New Outline Flowchart



Evaluation and Metrics

Metric	Preview	Draft3
Quality	Subjective listening	FAD (Fréchet Audio Distance)
Relevance	"Does it match?" (opinion)	SBERT similarity score
Coverage	7 test cases	Ready for 50–100 per emotion
Failure modes	Unclear	Documented (domain gap, threshold limits)



A Side Quest

Remember the video by Andrej Karpathy we spoke about in the first presentation. Just to try it out in a similar GPT-2 manner, I trained a musical notes generation model from scratch. I found a dataset with ABC notation as input and outputs musical notes in the form of ABC. We can test it using a website which plays the ABC notes.

```
# generate from the model
context = torch.zeros(1, 1, dtype=torch.long, device=device)
print(decode_n_generate(context, max_new_tokens=2000)[0].tolist())

'''
B_2 D | CDE G_2 F | G_2 F : c_2 c | B_2 D EGB | F_2 F_2 F | G_2 EDB, | G_A B, B_6 G, |
def B_2 | AAA FA | GFE EDB, | G_2 E DB_B, | def B_2 c | def B_2 :

X:161282
L:1/8
M:1/2

X:"Mw" a | fed B2 c | | d2 e f2 g | a7fjge | fed F2 {dc} | d2 d2 d2 : f2 d e2 f | g2 |A|AAA |Acde c2 A |
Ac2 fce | a2 e2 g | f2 f2 A | {ggf} f2 | f1 | agc c2 f | fdc ffc |
B2 TC cBA | {G32 G}c{f2} G3 A | ccc ccc f {G3} - a2 c2 a | {g2e} g2 |
{cfr} {2r cdb | {B} a2 C G4 | A G F | F1 B2 A | {G3 c} f2 | {d} {c3 c} A2 : |A/B/|AD| {B} |
{FA}B8 | {A/G/}A/G/ {F2} D | E2 G | {DA} .BA/G/ A/ | - |g{c/r}A/G/ d | {G3,G/FG/ A/ | G | {B/c/} B/A/ |
{G/A}B8 c | G G D | G A | B A {C2 e}d | {g2c/}B/A/ A | {E} D | {G} {E} | G D F | G A c | d c/B/ A |
B G | G | {A} A | A C G4 | A G F | F1 B2 : |A/B/|Mw" a2 |
~ ~ ~ |A| F | B | e2 c {d/r/} T T | e | {f/r/a} z | e "A" {f} |
{f d} e | {d c} | {f g} {ag} f1 e2 {d d} | {c c} c | {ag} c | a2 {eg} | {f e} d2 |
{f g} {F= E} AB | {d} {Gc/} {B} | {ag} {f g} d2 {e f} | {g f} d2 {f e} | b3/2 {ag} {f e} {c} | {c' b} {a d} - | gec' a |
{c f} g2 | {f r/a} {d c} {d} - {c} | {A G} {e} {f d} {f g} | {c c} {A G} {d A} {G c} {d A} |
d2 {AG2} {e2} | {d} {Gc/} {B} | {ag} {f g} | {c d} {e} {f} B2 | {g2} f2 g | {AG} {d2 c' b} {e} A2 G : |
```

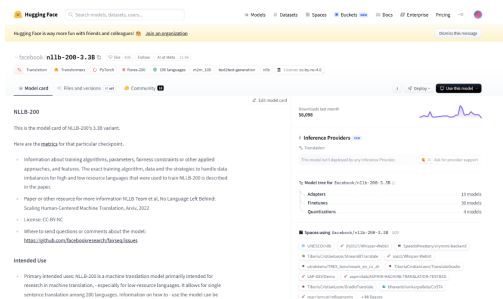
For more information, see [the project page](#).

```
X: 1
T: Cooley's
M: 4/4
L: 1/8
R: reel
K: Emin
|:D2|E|B|c|B A B2 E|~B2 A B dBA|FDAD BDAD|FDAD dAFD|
EBBA B2 E|B|B2 AB dfg|afe~c dBA|f|DEFD E2:|
|:gfe|B B2 efge|B B2 ged|B|A2 FA dAFA|A2 FA dfg|
eB A2 eB|B|B B2 dfg|afe~c dBA|f|DEFD E2:|
```

Multilingual Support for Wider Accessibility

- Integrated a **language translation model** for multilingual input.
- Users interact in **different languages**, not just English.
- Output remains **consistent with input language**.
- Improves usability for:
 - elderly users
 - single-language users
 - global users

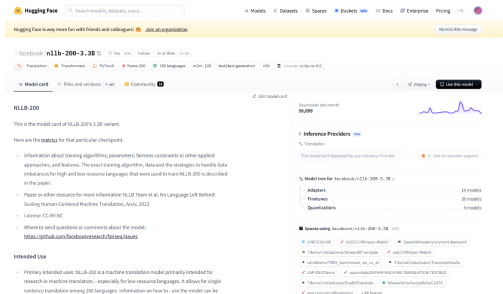
Model: huggingface.co/facebook/nllb-200-3.3B



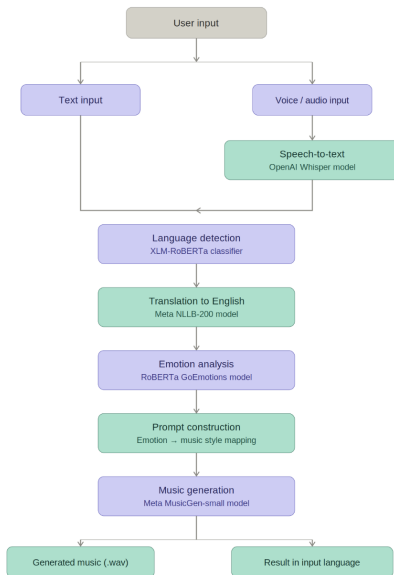
Expanded Input Modes and UI Improvement

- Added new UI functions for:
 - recording-based voice input
 - source audio analysis
- The system is no longer limited to typed text.
- Users can now generate music through:
 - text input
 - spoken language input
 - uploaded audio input
- Added a **transcription display function** to show recognised audio directly.
- Improves inclusivity for users with different needs.

Model: huggingface.co/openai/whisper-base



System Pipeline



Cinematic Scene-to-Score Approach

Traditional AI music systems usually generate audio only, which makes the result difficult to edit, reuse, or develop further in a real composition workflow.

My approach extends this idea into a composer inspiration tool:

- Interpret text as a **cinematic scene**, not just an emotion
- Generate **film-style music** using structured prompts
- Produce both:
 - High-quality **audio** representing the soundtrack of the scene
 - **Symbolic music (MIDI + score)** using music21, for editing and reuse

Goal: To create a system that helps musicians by generating music that can be listened to, edited, and developed into a real soundtrack.

Key Technical Additions and System Design

Cinematic knowledge layer

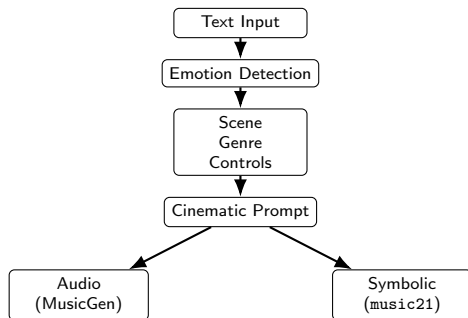
- Scene classification (romance, suspense, action, etc.)
- Scene arcs to control musical evolution (build-up, climax, resolution)
- Leitmotifs to represent narrative roles (hero, villain, memory, etc)
- Genre-based instrumentation (drama, horror, sci-fi, etc.)

User controls

- Genre, intensity, creativity and leitmotif selection
- Ending style (fade, resolved, unresolved, sudden stop)

Symbolic generation with music21

- Emotion → tempo, key, mode, instrument
- Multi-track structure: melody, harmony, bass
- MIDI + score output for editing

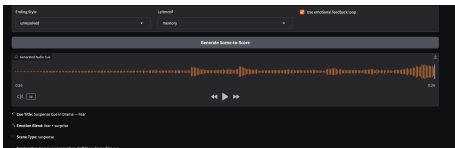


Results: Audio and Symbolic Output

Input scene (inspired by the emotional narrative of Bohemian Rhapsody):

“Is this the real life, or is this just fantasy? A young man feels trapped between reality and illusion after making a life-changing mistake, overwhelmed by guilt and fear.”

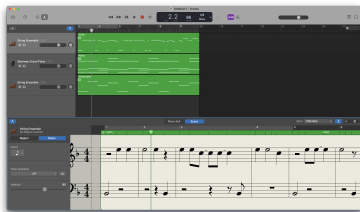
Audio output (MusicGen)



Realistic cinematic soundtrack generated in the waveform domain.

[Listen to generated audio](#)

Symbolic output (MIDI + score using music21)



Editable musical draft exported as MIDI and score for composition.

Important observation:

The MIDI output is not identical to the generated audio. This is because the audio is produced by a neural model in the waveform domain, while the MIDI is a simplified symbolic representation designed for composition.

Key insight:

The audio provides realism, while the symbolic score provides structure, enabling further musical development.

Contributions

Ishan	Haoran	Alba
Outline (Presentation 1)	Slides (Presentation 1)	Beamer template
Outline (Presentation 2 & 3)	Slides (Presentation 2 & 3)	Slides (Final Presentation)
Demo notebook shown during progress ppt	Created GitHub repository	Pushed the notebook on GitHub
Emotion Conditioned Music Generation with Semantic Alignment (Colab notebook)	Multilingual model, expanded input modes and richer emotion model (Colab notebook)	Multimodal Cinematic Scene-to-Score Generation with Audio and Symbolic Representation (Colab notebook)

Thank you

... Because Your Emotions Deserve a Soundtrack!!